PAPER<i>418 – F</i>U Sun: Complete SCm Solar Cycle Final Calibration with All Five Components

Session: *110 (grokshare755feea7.txt analysis)*

PAPER*418 – F*U Sun: Complete SCm Solar Cycle Final Calibration with All Five Components

Source: grok*share*755feea7.txt — "Star Magic" Final Calibration Section + Full FU derivation **Session:** 110 (grok*share*755feea7.txt analysis) **CP4 Class:** `FUSunCompleteSCmSolarCycleFinalCalibrationCalculator` (#68)

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1. Overview

PAPER_418 presents the **complete numerically calibrated F_U** for the Sun incorporating all five UQFF components (Ug1, Ug2, Ug3, Um, $A_{\mu\nu}$) with the full 11-year solar cycle modulation. This is the synthesis paper integrating PAPER_409–417 into a single predictive formula. All five coupling constants are finalized: $k_1=1.5$, $k_2=1.2$, $k_3=1.8$, $\beta=0.6$, $\eta=10^{-22}$.

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2. Complete FU Master Equation

```
F_U = \underbrace{\sum_{i=1}^{4} k_i \cdot Ug_i}_{\text{gravitational}} -
\underbrace{\sum_{i=1}^{4} \beta_i \cdot Ug_i \cdot \frac{\Omega_g M_{\text{BH}}}{d_g}
\cdot E_{\text{react}}}_{\text{buoyancy}} + \underbrace{U_m}_{\text{magnetic}} +
\underbrace{A_{\mu\nu}}_{\text{spacetime}}
```

In expanded tensor notation:

```
F_U = \sum_i \left[k_i \cdot Ug_i - \beta_i \cdot Ug_i \cdot \frac{\Omega_g
M_{\text{BH}}}{d_g} \cdot E_{\text{react}}\right] + \sum_j
\frac{\mu_j(t,[\text{SCm}])}{r_j}\left(1 - e^{-\gamma t \cos(\pi t_n)}\right)\hat{\phi}_j
+ (g_{\mu\nu} + \eta T_s^{\mu\nu})
```

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3. Per-Component Solar Calibration

3.1 Component 1: Ug1 (DPM Surface Term)

```
Ug_1(t) = 1.5 \cdot \mu_{s,\text{full}}(t) \cdot 274 \cdot e^{-\alpha t} \cdot \cos(\pi
t_n) \cdot (1 + 0.01\sin(0.001t))
```

With `\mu_{s,\text{full}}(t) = 3.38 \times 10^{23} + \delta_\mu \sin(\omega_c t)`:

```
Ug_1(t=0) \approx 1.5 \times 3.38 \times 10^{23} \times 274 \approx 1.39 \times 10^{26}
```

Solar cycle amplitude:

```
\delta_{Ug1} = 1.5 \times \underbrace{4 \times 10^{23}}_{\delta_\mu} \times 274 \approx
1.64 \times 10^{26} \times 0.01 \approx 4.68 \times 10^{24}
```

where the `4 \times 10^{23} \frac{\Delta B_s}{\Delta B_{\text{SCm}}} R_s^3 \approx 4 \times 10^{-5} \times 3.38 \times 10^{20}/(10^{-4}) \approx` varies with `\sin(\omega_c` t).

Simplified Ug1 term in F_U:

$$F_{U,\text{Ug1}} \approx (1.17 \times 10^{27} + 4.68 \times 10^{24} \cdot \sin(\omega_c t)) \cdot e^{-0.001t} \cdot \cos(\pi t)$$

3.2 Component 2: Ug2 (Heliosphere Bubble)

$$Ug_2 = 1.2 \cdot (\rho_{\text{vac,UA}} + \rho_{\text{vac,SCm}}) \cdot \frac{M_s}{r^2} \cdot S(r-R_b) \cdot (1 + \delta_{sw} v_{sw}) \cdot H_{\text{SCm}} \cdot E_{\text{react}}$$

With $H_{\text{SCm}} \approx 1$ and $E_{\text{react}} \approx 10^{54} e^{-0.0005t}$:

$$Ug_2 \approx 1.2 \times (10^{-23} + 10^{-30}) \times \frac{2 \times 10^{30}}{(1.5 \times 10^{11})^2} \times 10^{54} \approx 9.83 \times 10^6$$

Simplified Ug2 term:

$$F_{U,\text{Ug2}} \approx 1.18 \times 10^{53} \cdot e^{-0.0005t}$$

3.3 Component 3: Ug3 (Core Disk)

$$Ug_3 = 1.8 \cdot B_j(t) \cdot \cos(\omega_s(t) \cdot t \cdot \pi) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

With $B_j(t) = (10^3 + 0.4\sin(\omega_c t))$ T, $P_{\text{core}} = 10^{-3}$:

$$Ug_3 \approx 1.8 \cdot (10^3 + 0.4\sin(\omega_c t)) \cdot \cos(\omega_s(t) \cdot t \cdot \pi) \cdot e^{-0.0005t}$$

Simplified Ug3 term:

$$F_{U,\text{Ug3}} \approx (1.8 \times 10^3 + 0.72 \cdot \sin(\omega_c t)) \cdot \cos(\omega_s(t) \cdot t \cdot \pi) \cdot e^{-0.0005t}$$

A more compact form:

$$F_{U,\text{Ug3}} \approx (1.005 \times 10^3 + 2.405 \cdot \sin(\omega_c t)) \cdot \cos((2.5 \times 10^{-6} - 0.4 \times 10^{-6} \sin(\omega_c t)) \cdot t \cdot \pi) \cdot e^{-0.0005t}$$

3.4 Component 4: Um (Magnetic Strings, $j = 10^9$ strands)

$$Um = \sum_{j=1}^{10^9} \frac{\mu_j(t,[\text{SCm}])}{r_j} \cdot (1 - e^{-0.0001t})$$

With $\mu_j = (B_s + B_\text{SCm})R_s^3 / N_j$:

$$Um \approx (2.26 \times 10^{19} + 9.04 \times 10^{16} \cdot \sin(\omega_c t)) \cdot (1 - e^{-0.0001t})$$

Simplified Um term:

$$F_{U,Um} \approx (2.26 \times 10^{19} + 9.04 \times 10^{16} \cdot \sin(\omega_c t)) \cdot (1 - e^{-0.0001t})$$

3.5 Component 5: $A_{\mu\nu}$ (Spacetime Metric)

$$A_{\mu\nu} \approx \underbrace{[1, -1, -1, -1]}_{\text{Minkowski}} + \underbrace{1.27 \times 10^{-20}}_{\eta \cdot T_s^{00}(\text{stellar})} + \underbrace{1.11 \times 10^{-16}}_{\eta \cdot T_s^{00}(\text{SCm})}$$

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4. Complete Simplified F_U (Sun)

```
\boxed{F_U \approx (1.17 \times 10^{27} + 4.68 \times 10^{24} \sin(\omega_c t)) \cdot e^{-0.001t} \cdot \cos(\pi t) \cdot (1 + 0.01\sin(0.001t))}
```

```
\quad + \; 1.18 \times 10^{53} \cdot e^{-0.0005t}
```

```
\quad + \; (1.005 \times 10^3 + 2.405\sin(\omega_c t)) \cdot \cos\!\left((2.5\times10^{-6} - 0.4\times10^{-6}\sin(\omega_c t)) \cdot t\pi\right) \cdot e^{-0.0005t}
```

```
\quad + \; (2.26\times10^{19} + 9.04\times10^{16}\sin(\omega_c t)) \cdot (1 - e^{-0.0001t})
```

```
\quad + \; [1,-1,-1,-1] + 1.27\times10^{-20} + 1.11\times10^{-16}
```

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5. Calibrated Constants Summary

Constant	Value	Source
k_1	1.5	Ug1 DPM surface coupling (PAPER_411)
k_2	1.2	Ug2 heliosphere bubble (PAPER_400)
k_3	1.8	Ug3 magnetic strings disk (PAPER_401)
k_4	2.0	Ug4 vacuum–BH (PAPER_402)
\beta_i	0.6	Buoyancy coupling universal (PAPER_403)
\eta	10^{-22}	Metric tensor coupling
\kappa	5 \times 10^{-4} day^{-1}	SCm decay rate
\omega_c	2\pi / (3.96 \times 10^8) s^{-1}	11-year solar cycle

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6. Code: Final FU Sun Computation

```
double compute_FU_sun_complete(double t, double tn) {
// Solar parameters
double Ms = 1.989e30, Rs = 6.96e8;
double Bs = 1e-4 + 0.4e-4 * sin(omega_c * t);
double BSCm = 1e3;
double mu_s = (Bs + BSCm) * pow(Rs, 3);
double grad = 274.0;
double delta_def = 0.01 * sin(0.001 * t);
double Ereact = 1e54 * exp(-0.0005 * t);
double omega_s = 2.5e-6 - 0.4e-6 * sin(omega_c * t);

// Five terms
double Ug1_term = 1.5 * mu_s * grad * exp(-0.001 * t) * cos(M_PI * tn) * (1.0 + delta_def);
double Ug2_term = 1.18e53 * exp(-0.0005 * t);
double Ug3_term = 1.8 * (1e3 + 0.4 * sin(omega_c * t)) * cos(omega_s * t * M_PI)
* 1e-3 * Ereact;
double Um_term = (2.26e19 + 9.04e16 * sin(omega_c * t)) * (1.0 - exp(-0.0001 * t));
double A_term = 1.0 + 1.27e-20 + 1.11e-16; // metric (00 component)

return Ug1_term + Ug2_term + Ug3_term + Um_term + A_term;
}
```

---

7. Unit Test

```
import math
def test_FU_sun_dominant_term():
"""Ug1 term dominates at t=0, tn=0"""
k1 = 1.5; mu_s_full = 3.38e23; grad = 274.0
FU_Ug1 = k1 * mu_s_full * grad * 1.0 * 1.0 # t=0, tn=0
assert 1e26 < FU_Ug1 < 1e28, f"FU_Ug1 out of expected range: {FU_Ug1}"
```
